

Elisa Valente.



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I build machine-learning systems for real-world decisions, and the data, simulation and evaluation work that makes them trustworthy.

Currently R&D Tech Lead at Bandora Systems, Portugal

Email

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Interactive portfolio

mariaevalente.github.io

Based in

Viana do Castelo, Portugal

About

I'm a computer engineer. I studied at the University of Minho, specialising in data science and intelligent systems, and did my master's research at UCL on adversarial deep learning for network anomaly detection.

I joined Bandora in 2021 as a data scientist and have led its R&D since May 2026. My work sits between machine learning, building physics and software engineering, and I like carrying an idea from the first experiment to something a client uses.

I also set up the team's engineering standards, mentor interns and master's students, and give talks inside and outside the company.

Outside work I travel with a camera, and I write and sing as Lara Blue.

2015–2020

BSc and MSc, Computer Science and Engineering, University of Minho

2020

Master's research, University College London

2021

Data Scientist, Bandora Systems

2024

Patent priority filed. Speaker at Ai4, Las Vegas

2026

R&D Tech Lead. Patent published. Speaker at AI Nest

What I bring

Ideas taken all the way, from the first experiment to something a client uses.

Evaluation I can defend: baselines, measurement and verification, and honest comparisons.

Tools other people use: a language for control rules and a reporting platform.

Clear communication across teams: I explain technical work to people without a technical background, so they understand it and can act on it, whether in a meeting, a client report or the documentation.

1st inventor

of patent application WO 2026/033501, HVAC control with models pre-trained on digital-twin data

–26%

HVAC energy of my predictive controller against fixed schedules, ten simulated buildings

10 of 10

buildings more comfortable than with the rule-based controller, with slightly less energy

Ai4 2024

speaker in Las Vegas on synthetic data for building control, and at AI Nest 2026

Talks



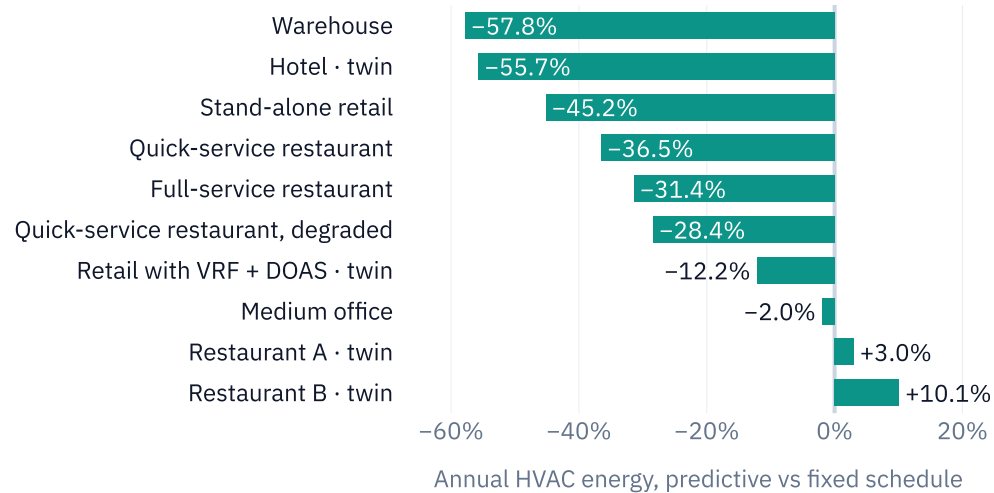
Ai4 2024, Las Vegas



AI Nest 2026, Coimbra

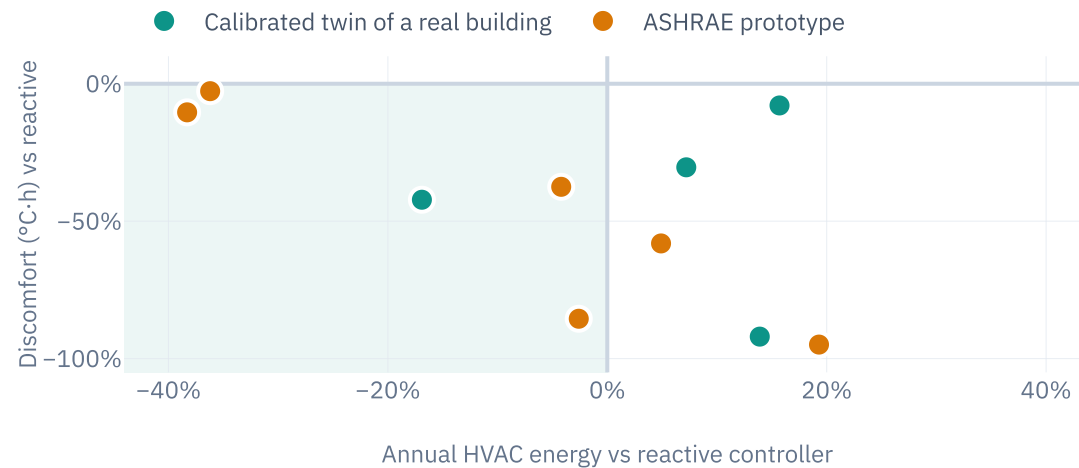
Energy against the fixed schedule

Annual HVAC energy change per building. Negative is a saving.



Comfort and energy against the rule-based controller

Each dot is a building. The shaded quarter is less energy and less discomfort.



Research, 2025–2026

Predictive HVAC control

A controller that forecasts every zone 30 minutes ahead and, every 15 minutes, picks the least energy-intensive action that keeps the occupied zones comfortable.

I tested it on ten simulated buildings, six ASHRAE prototypes and four digital twins of real buildings, over a full year. Each building ran under a fixed schedule, a rule-based controller and the predictive controller, with the same weather, occupancy and constraints.

-25.6%

energy against the fixed schedule

-46%

discomfort against the rule-based controller

6.4×

fewer candidate actions per decision in the second generation

0.35 °C

closed-loop forecast error, 30 minutes ahead

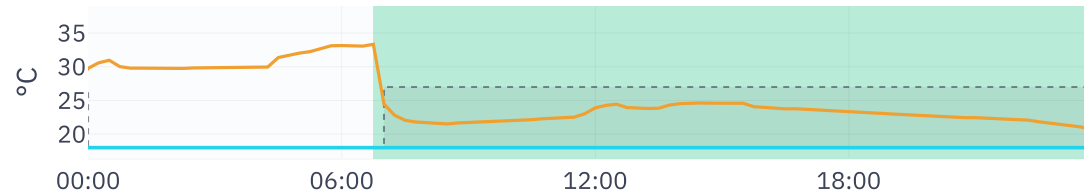
Role Technical lead and main developer. The decision method is proprietary and is shown only through its results.

— Room temperature — Setpoint Comfort range, during opening hours Running · cooling
 Running · heating Running · fan only Off (year strip)

Predictive controller (Arrow v2)

Building HVAC energy **34.3 kWh** Zone discomfort **0.0 °C·h** In comfort band **100%** of occupied time

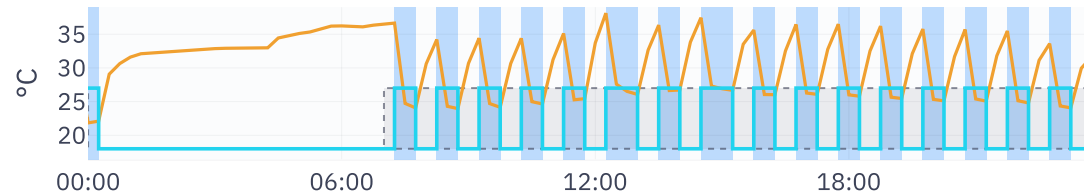
Zone equipment running **17.0 h**



Reactive rule-based controller

Building HVAC energy **28.6 kWh** Zone discomfort **58.6 °C·h** In comfort band **47%** of occupied time

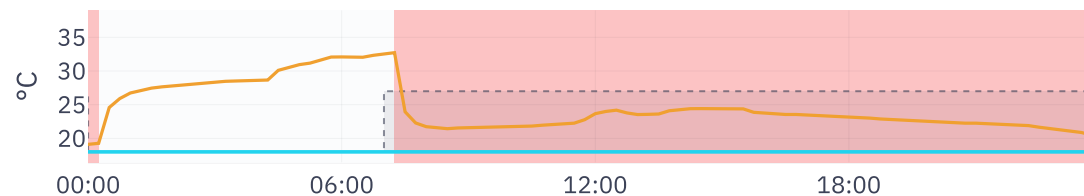
Zone equipment running **9.0 h**



Fixed schedule

Building HVAC energy **73.6 kWh** Zone discomfort **2.8 °C·h** In comfort band **97%** of occupied time

Zone equipment running **17.0 h**



Visual analytics

The Valente chart

The view I designed to show what the equipment did and how the room responded, on one time axis: running periods coloured by mode, the comfort range during opening hours, the setpoint and the room temperature. It closes every monthly client report.

Here, 5 November in a simulated restaurant kitchen. The predictive controller holds the range with the fan alone. The fixed schedule heats all day for the same temperatures at about twice the energy. The rule-based controller cycles the cooling and leaves the kitchen above range for half its opening hours.

Online A full year for three buildings, day by day, at mariaevalente.github.io.

Quick-service restaurant (simulated)

ASHRAE 90.1 prototype, degraded · EnergyPlus · Lisbon weather
April 2021 – June 2021

Summary

COMFORT HELD 95%	CONSUMPTION 5,625 kWh	COST —	IN MANUAL 0%
SAVED 2,025 kWh		SAVED —	

Counted over the 3 months with complete data; any month still in progress is listed below but not included.

Consumption and cost



MONTH	USED (KWH)	COST (€)	EXPECTED (KWH)	SAVED (KWH)	SAVED (%)	SAVED (€)	CONSUMPTION
Apr 2021	1,244	—	2,298	1,054	46%	—	complete
May 2021	1,836	—	2,519	684	27%	—	complete
Jun 2021	2,545	—	2,833	288	10%	—	complete

The expected figures are the same simulated building, with the same weather, occupancy and comfort ranges, run on a fixed time schedule instead of the predictive controller.

Product engineering, 2026

Analytics and reporting platform

A Streamlit application over the company's asset model (PostgreSQL) and time-series store (MongoDB). For a building or a portfolio it shows what the controller did, the comfort it held, what was consumed and saved, and how complete the data is. Every month it writes a single HTML report for the client.

The report shown runs the production renderer on the simulated restaurant, April to June, against the same building on a fixed schedule.

13 s → 3 ms

filtered time-series read after the index work

4× faster

monthly report: under 14 seconds instead of almost a minute

5 pages

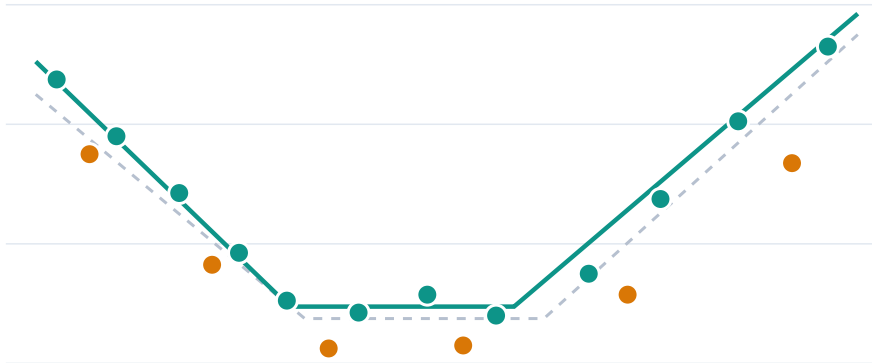
about 14,000 lines of Python

Docker

multi-stage, non-root, behind an authenticating proxy

Role Sole developer.

Savings benchmark



When a contract is paid from savings, the baseline is what gets invoiced. I set up the climate-corrected baseline and supervised its move to a tournament of ASHRAE Guideline 14 change-point models: the most accurate one meeting $CV(RMSE) \leq 15\%$ and $|NMBE| \leq 5\%$ becomes the benchmark.

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buildings with a compliant model

5.6%

median $CV(RMSE)$ of the models selected

Role Method design and supervision. Implemented by a building analyst.

Patent and synthetic data

WO 2026/033501 A1

Method for controlling HVAC equipment of a building, computer system and computer program product thereof

Inventors **Maria Elisa Maciel Valente**, R. Teixeira, G. Karafotias, M. Pereira

Priority 9 August 2024

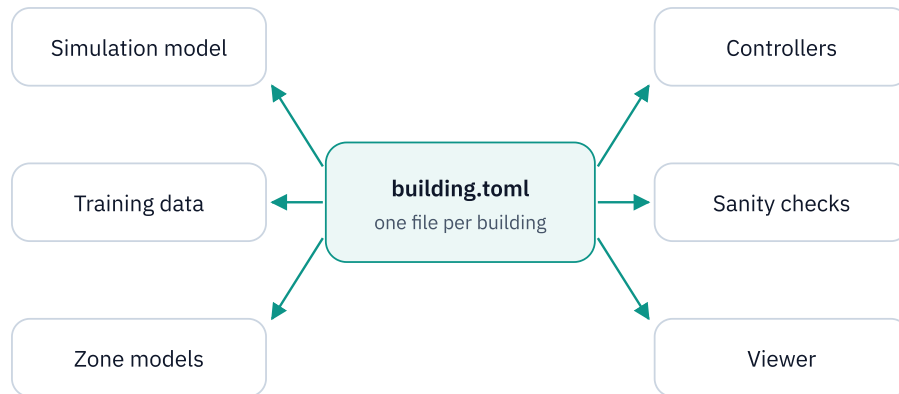
Published 12 February 2026

A new building has no useful history to learn from. The method pre-trains the energy and temperature forecasters on synthetic data from the building's digital twin, combined with whatever was measured, so that control works from the first day.

It grew out of a 2022 warm-up optimiser, XGBoost forecasts searched with simulated annealing, that cut HVAC cost inside the morning warm-up window by 27 to 45% in simulation. I presented the approach at Ai4 2024 in Las Vegas and at AI Nest 2026 in Coimbra.

Role First inventor.

A simulation platform for any building



One TOML file describes a building: zones, comfort ranges, opening hours, equipment and its command map. Simulation, data generation, training, the controllers and the checks all read it, so a new building is a new file instead of new code. 71 buildings and archetypes use the format. I also designed the team's reproducibility workflow with DVC, MLflow and Git.

Role Spec schema, API layer and sanity tests. Testbed built with the R&D team.

BuildFlow

```

if comfort_less([1, 2]) > 0:
    ACTUATION([1], [{"Mode", "Cooling"}], [])
else:
    if comfort_less([1, 2]) < 0:
        ACTUATION([1], [{"Mode", "Heating"}], [])
    else:
        ACTUATION([1], [{"Mode", "Fan"}], []);
  
```

A small domain-specific language for expert control rules, with its own PLY parser, a compiler to JSON and an interpreter, designed so that engineers could write and review rules without touching code. I wrote the grammar, the compiler, the interpreter and their documentation.

Role Designer and developer.

More at mariaevalente.github.io

The explorable Valente chart, the full client report and the benchmark explainer. mariaemvalente13@gmail.com